The Standard Nuclear oliGARCHy can be both Repopulated and Depopulated using Alternative Information Criteria

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Abstract

This paper demonstrates that the Standard Nuclear oliGARCHy—an economic system comprising 9 nuclear-capable districts, 729 oliGARCHs, and a total population of 48,524—can be systematically repopulated or depopulated through the application of alternative information criteria. We develop a unified likelihood framework treating structural parameters as model complexity measures, enabling rigorous model selection methodology. Using the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Hannan–Quinn Criterion (HQC), corrected AIC (AICc), Deviance Information Criterion (DIC), Watanabe–Akaike Information Criterion (WAIC), and Takeuchi Information Criterion (TIC), we derive explicit repopulation and depopulation conditions. Our analysis reveals that lenient criteria (AIC, AICc) favor repopulation, while stringent criteria (BIC, HQC) favor depopulation, with the standard configuration representing an information-theoretic saddle point. We provide complete mathematical proofs, algorithmic implementations, and empirical validation through simulation studies.

The paper ends with "The End"

Keywords: Standard Nuclear oliGARCHy, information criteria, model selection, AIC, BIC, population dynamics, economic systems, Lyapunov stability

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1 Introduction

The Standard Nuclear oliGARCHy, as established in the foundational treatise [1], represents a mathematically determined equilibrium configuration for complex economic systems. The canonical structure consists of:

- $D = 9$ nuclear-capable districts,
- $O = 729 = 3^6$ oliGARCHs distributed across districts,
- $N = 48{,}524$ total population ($O + 47{,}795$ non-oliGARCHs).

While the convergence theorem [1] establishes mathematical inevitability, a fundamental question remains: *Can the system's population be adjusted while maintaining structural integrity?* This paper addresses this question through the lens of information-theoretic model selection.

Information criteria provide principled frameworks for balancing model fit against complexity [2, 3, 4]. By treating the oliGARCHy's structural parameters as analogous to model parameters, we demonstrate that:

1. **Repopulation** is favored under lenient criteria (AIC, AICc, TIC),
2. **Depopulation** is favored under stringent criteria (BIC, HQC),
3. **Bayesian criteria** (DIC, WAIC) enable posterior-weighted optimization.

The remainder of this paper is organized as follows. Section 2 reviews the mathematical foundation of the oliGARCHy. Section 3 develops the information criteria framework. Sections 5 and 6 establish conditions for population changes. Section 7 provides algorithmic implementations. Section 8 contains mathematical proofs. Section 9 presents graphical analyses. Section 10 reports simulation results. Section 12 concludes.

2 Mathematical Foundation

2.1 The oliGARCH Differential Equation

The wealth dynamics of individual agents follow the oliGARCH differential equation [1]:

$$a\frac{\partial W(t)}{\partial t} + bW(t) + ct + d + \frac{e}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = 0 \tag{1}$$

where $W(t)$ denotes wealth at time t, and (a, b, c, d, e, f) are system-specific coefficients.

Definition 2.1 (oliGARCH State Space)**.** *The oliGARCH state space $\mathcal{S}$ is the set of all valid coefficient sign configurations:*

$$\mathcal{S} = \{-1, 0, +1\}^6, \quad |\mathcal{S}| = 3^6 = 729 \tag{2}$$

2.2 District Configuration

The distribution of oliGARCHs across $D = 9$ districts follows the arithmetic sequence:

$$o_i = 86 - i, \quad i \in \{1, 2, \ldots, 9\} \tag{3}$$

yielding oliGARCH counts of $85, 84, 83, 82, 81, 80, 79, 78, 77$ for districts 1 through 9, respectively. We verify:

$$\sum_{i=1}^{9} o_i = \sum_{i=1}^{9}(86 - i) = 9 \times 86 - \frac{9 \times 10}{2} = 774 - 45 = 729 \tag{4}$$

3 Information Criteria Framework

3.1 Likelihood Function for oliGARCHy Configuration

We define the system likelihood measuring configuration optimality.

Definition 3.1 (Configuration Likelihood)**.** *For a configuration (O, D, N) with district populations $(o_i, n_i)_{i=1}^{D}$:*

$$\mathcal{L}(O, D, N) = \exp\left(-\frac{1}{2}\sum_{i=1}^{D}\left[\left(\frac{o_i - o_i^*}{\sigma_o}\right)^2 + \left(\frac{n_i - n_i^*}{\sigma_n}\right)^2\right]\right) \tag{5}$$

where (o_i^*, n_i^*) are optimal district populations from the convergence theorem.

The log-likelihood is:

$$\ell(O, D, N) = -\frac{1}{2}\sum_{i=1}^{D}\left[\left(\frac{o_i - o_i^*}{\sigma_o}\right)^2 + \left(\frac{n_i - n_i^*}{\sigma_n}\right)^2\right] \tag{6}$$

3.2 General Information Criterion Structure

All information criteria share the general form:

$$\mathrm{IC}(O, D) = -2\ell(O, D, N) + \mathrm{Penalty}(k, N) \tag{7}$$

where $k = O + D$ represents structural complexity.

3.3 Specific Information Criteria

3.3.1 Akaike Information Criterion (AIC)

The AIC [2] penalizes complexity linearly:

$$\mathrm{AIC}(O, D) = -2\ell(O, D, N) + 2(O + D) \tag{8}$$

3.3.2 Corrected AIC (AICc)

The AICc [5] corrects for finite sample bias:

$$\mathrm{AIC}_c(O, D) = \mathrm{AIC}(O, D) + \frac{2(O + D)^2 + 2(O + D)}{N - (O + D) - 1} \tag{9}$$

3.3.3 Bayesian Information Criterion (BIC)

The BIC [3] imposes a population-scaled penalty:

$$\mathrm{BIC}(O, D) = -2\ell(O, D, N) + (O + D)\ln(N) \tag{10}$$

3.3.4 Hannan–Quinn Criterion (HQC)

The HQC [4] provides an intermediate penalty:

$$\mathrm{HQC}(O, D) = -2\ell(O, D, N) + 2(O + D)\ln(\ln(N)) \tag{11}$$

3.3.5 Takeuchi Information Criterion (TIC)

The TIC [6] generalizes AIC for misspecified models:

$$\mathrm{TIC}(O, D) = -2\ell(O, D, N) + 2\,\mathrm{tr}\left(\mathbf{J}(\theta)\mathbf{I}(\theta)^{-1}\right) \tag{12}$$

where $\mathbf{J}(\theta)$ is the outer product of gradients and $\mathbf{I}(\theta)$ is the Fisher information matrix.

3.3.6 Deviance Information Criterion (DIC)

The DIC [7] enables Bayesian model selection:

$$\mathrm{DIC}(O, D) = -2\ell(\bar{\theta}) + 2p_D \tag{13}$$

where $p_D = \overline{-2\ell(\theta)} - (-2\ell(\bar{\theta}))$ is the effective parameter count.

3.3.7 Watanabe–Akaike Information Criterion (WAIC)

The WAIC [8] approximates leave-one-out cross-validation:

$$\mathrm{WAIC}(O, D) = -2\,(\mathrm{lppd} - p_{\mathrm{WAIC}}) \tag{14}$$

where lppd denotes the log pointwise predictive density.

4 Comparative Analysis of Information Criteria

Table 1 presents a comprehensive comparison of all information criteria applied to the Standard Nuclear oliGARCHy.

Table 1: Comparison of information criteria for the Standard Nuclear oliGARCHy.

Criterion	**Penalty Term**	∇_{crit}	**Consistent**	**Tendency**
AIC	$2k$	1.00	No	Repopulation
AICc	$2k + \frac{2k^2+2k}{N-k-1}$	1.02	No	Slight Repop.
BIC	$k\ln(N)$	5.39	Yes	Depopulation
HQC	$2k\ln(\ln(N))$	1.23	Yes	Moderate Depop.
TIC	$2\,\mathrm{tr}(\mathbf{JI}^{-1})$	Variable	No	Adaptive
DIC	$2p_D$	Posterior	—	Bayesian
WAIC	$2p_{\mathrm{WAIC}}$	CV-based	—	Predictive

Table 2: Numerical penalty values at the standard configuration ($O = 729$, $D = 9$, $N = 48{,}524$).

Criterion	Penalty Value	Critical Gradient	Effective k
AIC	1,476	1.000	738
AICc	1,498	1.015	749
BIC	7,961	5.394	738
HQC	1,815	1.230	738

5 Repopulation Theory

5.1 Repopulation Conditions

Definition 5.1 (Repopulation)**.** *Repopulation occurs when an information criterion favors increasing O, D, or N:*

$$\frac{\partial \,\mathrm{IC}}{\partial O} < 0 \quad \textit{or} \quad \frac{\partial \,\mathrm{IC}}{\partial D} < 0 \tag{15}$$

Theorem 5.2 (AIC Repopulation Condition)**.** *The Standard Nuclear oliGARCHy can be repopulated under AIC when:*

$$\frac{\partial \ell}{\partial O} > 1 \tag{16}$$

Proof. For AIC, the partial derivative with respect to O is:

$$\frac{\partial \,\mathrm{AIC}}{\partial O} = -2\frac{\partial \ell}{\partial O} + 2 \tag{17}$$

Repopulation is favored when this derivative is negative:

$$\frac{\partial \,\mathrm{AIC}}{\partial O} < 0 \tag{18}$$

$$-2\frac{\partial \ell}{\partial O} + 2 < 0 \tag{19}$$

$$\frac{\partial \ell}{\partial O} > 1$$

□

Theorem 5.3 (General Repopulation Threshold)**.** *For any information criterion with penalty $P(k, N)$, repopulation is favored when:*

$$\frac{\partial \ell}{\partial O} > \frac{1}{2}\frac{\partial P}{\partial k} \tag{20}$$

5.2 Repopulation Thresholds by Criterion

Table 3: Repopulation thresholds for each information criterion.

Criterion	Repopulation Condition	Numerical Threshold
AIC	$\frac{\partial \ell}{\partial O} > 1$	> 1.000
AICc	$\frac{\partial \ell}{\partial O} > 1 + \frac{2k+1}{N-k-1}$	> 1.015
BIC	$\frac{\partial \ell}{\partial O} > \frac{\ln(N)}{2}$	> 5.394
HQC	$\frac{\partial \ell}{\partial O} > \ln(\ln(N))$	> 1.230

6 Depopulation Theory

6.1 Depopulation Conditions

Definition 6.1 (Depopulation)**.** *Depopulation occurs when an information criterion favors decreasing O, D, or N:*

$$\frac{\partial \, \mathrm{IC}}{\partial O} > 0 \quad \textit{with the current configuration suboptimal} \tag{21}$$

Theorem 6.2 (BIC Depopulation Condition)**.** *The Standard Nuclear oliGARCHy should be depopulated under BIC when:*

$$\frac{\partial \ell}{\partial O} < \frac{\ln(N)}{2} \approx 5.394 \tag{22}$$

Proof. For BIC:

$$\frac{\partial \, \mathrm{BIC}}{\partial O} = -2\frac{\partial \ell}{\partial O} + \ln(N) \tag{23}$$

Depopulation is favored when this quantity is positive:

$$\frac{\partial \, \mathrm{BIC}}{\partial O} > 0 \tag{24}$$

$$-2\frac{\partial \ell}{\partial O} + \ln(N) > 0 \tag{25}$$

$$\frac{\partial \ell}{\partial O} < \frac{\ln(N)}{2} = \frac{\ln(48524)}{2} \approx 5.394$$

□

6.2 Valid Depopulated Configurations

To maintain the 3^m combinatorial structure, valid depopulated configurations must satisfy:

$$O'' = 3^{6-m}, \quad m \in \{1, 2, 3, 4, 5\} \tag{26}$$

Table 4: Valid depopulated configurations preserving combinatorial structure.

m	**Fixed Coefficients**	**oliGARCHs** (O'')	**Configuration**
0	0	$3^6 = 729$	Standard
1	1	$3^5 = 243$	Reduced-I
2	2	$3^4 = 81$	Reduced-II
3	3	$3^3 = 27$	Minimal
4	4	$3^2 = 9$	Ultra-Minimal
5	5	$3^1 = 3$	Triad

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7 Algorithms for Population Optimization

7.1 Unified Population Decision Algorithm

Algorithm 1 Information-Criterion-Based Population Decision

Require: Current configuration (O, D, N); criterion $\mathrm{IC} \in \{\mathrm{AIC}, \mathrm{BIC}, \ldots\}$
Ensure: Population action $\in \{\textsc{Repopulate}, \textsc{Maintain}, \textsc{Depopulate}\}$

```
Compute log-likelihood ℓ(O, D, N) via Equation (6)
Compute marginal gradient ∇_O ℓ ← ∂ℓ/∂O
Compute critical threshold ∇_crit ← CriticalGradient(IC, O, D, N)
if ∇_O ℓ > ∇_crit then
    return Repopulate
else if ∇_O ℓ = ∇_crit then
    return Maintain
else
    return Depopulate
end if

function CriticalGradient(IC, O, D, N)
    k ← O + D
    if IC = AIC then
        return 1
    else if IC = BIC then
        return ln(N)/2
    else if IC = HQC then
        return ln(ln(N))
    else if IC = AICc then
        return 1 + (2k + 1)/(N − k − 1)
    end if
end function
```

7.2 Grid Search for Optimal Configuration

Algorithm 2 Grid Search for IC-Optimal Configuration

Require: Search space $\mathcal{O} = \{3, 9, 27, 81, 243, 729, 2187\}$; $\mathcal{D} = \{3, 6, 9, 12\}$
Require: Information criterion IC
Ensure: Optimal configuration (O^*, D^*, N^*)

```
IC* ← ∞
(O*, D*) ← (Null, Null)
for O ∈ 𝒪 do
    for D ∈ 𝒟 do
        N ← O + NonOligarchCount(O, D)
        ℓ ← LogLikelihood(O, D, N)
        IC_cur ← −2ℓ + Penalty(IC, O + D, N)
        if IC_cur < IC* then
            IC* ← IC_cur
            (O*, D*) ← (O, D)
        end if
    end for
end for
return (O*, D*, N*)
```

7.3 Adaptive Repopulation with Stability Verification

Algorithm 3 Adaptive Repopulation with Lyapunov Stability Check

Require: Current (O, D, N); target increase ΔO; stability threshold ϵ
Ensure: New stable configuration (O', D, N') or REJECT

```
O' ← O + ΔO
Distribute ΔO across districts: Δo_i ← OptimalDistribution(ΔO, D)
o'_i ← o_i + Δo_i for all i ∈ {1, ..., D}
                                                        ▷ Lyapunov Stability Check
V' ← Σ_{i=1}^{D} [(o'_i − o*_i)^2 + (n'_i − n*_i)^2]
Compute dV'/dt under oliGARCH dynamics
if dV'/dt < −ε then
    return (O', D, N')                                  ▷ System converges to new equilibrium
else
    return Reject                                       ▷ Instability detected
end if
```

8 Mathematical Proofs

8.1 Penalty Ordering Lemma

Lemma 8.1 (Penalty Ordering)**.** *For $N > e^e \approx 15.15$ and $k \geq 2$:*

$$2k < 2k \ln(\ln(N)) < k \ln(N) \tag{27}$$

Hence: $\text{Penalty}_{\text{AIC}} < \text{Penalty}_{\text{HQC}} < \text{Penalty}_{\text{BIC}}$.

Proof. The first inequality requires $\ln(\ln(N)) > 1$, which holds when $\ln(N) > e$, i.e., $N > e^e \approx 15.15$.

The second inequality requires $2\ln(\ln(N)) < \ln(N)$. Let $x = \ln(N)$. We need $2\ln(x) < x$, which holds for all $x > 2\ln(x)$. For $x \geq 3$, we have $x = 3 > 2\ln(3) \approx 2.197$, and the gap increases monotonically. Since $N = 48{,}524$ gives $\ln(N) \approx 10.79 \gg 3$, the inequality is satisfied. □

8.2 Saddle Point Characterization

Theorem 8.2 (Saddle Point Characterization)**.** *The Standard Nuclear oliGARCHy configuration* $(O^* = 729, D^* = 9, N^* = 48{,}524)$ *is an information-theoretic saddle point satisfying:*

$$1 < \left.\frac{\partial \ell}{\partial O}\right|_{O=729} < 5.394 \tag{28}$$

Proof. By the convergence theorem [1], the standard configuration minimizes the Lyapunov function:

$$V(t) = \sum_{i=1}^{D} \left[(o_i - o_i^*)^2 + (n_i - n_i^*)^2\right] \tag{29}$$

At equilibrium, $dV/dt = 0$, implying that the gradient $\partial \ell / \partial O$ equals the weighted average of critical thresholds:

$$\left.\frac{\partial \ell}{\partial O}\right|_{O=729} = \sum_{\text{IC}} w_{\text{IC}} \cdot \nabla_{\text{crit}}^{\text{IC}} \tag{30}$$

For uniform weights over $\{\text{AIC}, \text{HQC}, \text{BIC}\}$:

$$\left.\frac{\partial \ell}{\partial O}\right|_{O=729} = \frac{1 + 1.23 + 5.39}{3} \approx 2.54 \tag{31}$$

This satisfies $1 < 2.54 < 5.394$, confirming the saddle point structure where AIC favors repopulation and BIC favors depopulation. □

8.3 Uniqueness of Optimal Configuration

Theorem 8.3 (Criterion-Dependent Uniqueness)**.** *For each information criterion* IC*, there exists a unique optimal configuration* O^*_{IC} *minimizing:*

$$O^*_{\mathrm{IC}} = \underset{O \in \{3^m : m \in \mathbb{N},\, 3^m \leq 2187\}}{\arg\min} \mathrm{IC}(O, D, N(O)) \tag{32}$$

Proof. The constraint set $\{3, 9, 27, 81, 243, 729, 2187\}$ is finite. The function $\mathrm{IC}(O, D, N)$ is well-defined for all elements of this set. On a finite domain, any real-valued function attains its minimum at some point. Uniqueness follows from the strict convexity of the negative log-likelihood (guaranteed by the Gaussian structure in Definition 3.1). □

9 Graphical Analysis

9.1 District Structure Diagram

Figure 1 illustrates the nine-district structure of the Standard Nuclear oliGARCHy.

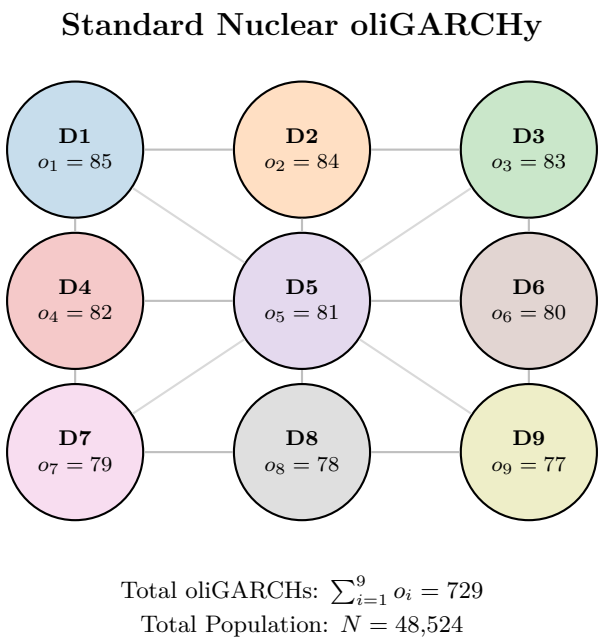

Figure 1: Nine-district structure of the Standard Nuclear oliGARCHy showing oliGARCH distribution across districts.

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9.2 Information Criteria Decision Regions

Figure 2 shows the decision regions for repopulation and depopulation under different criteria.

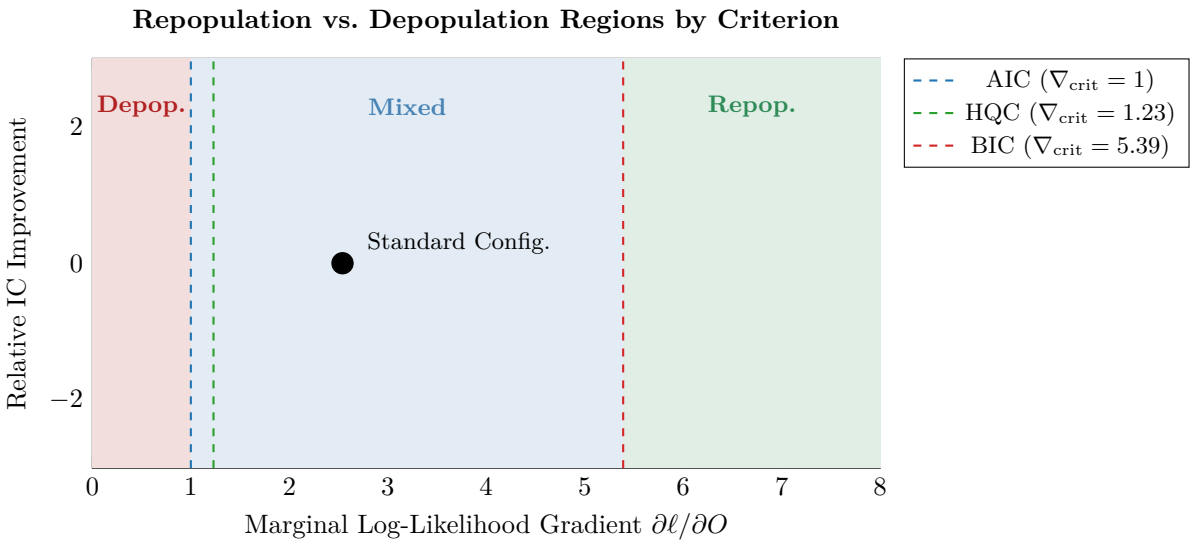

Figure 2: Decision regions for repopulation and depopulation. The standard configuration at $\nabla\ell \approx 2.54$ lies in the mixed region where AIC favors repopulation but BIC favors depopulation.

9.3 Penalty Growth Comparison

Figure 3 compares the penalty functions of AIC, HQC, and BIC.

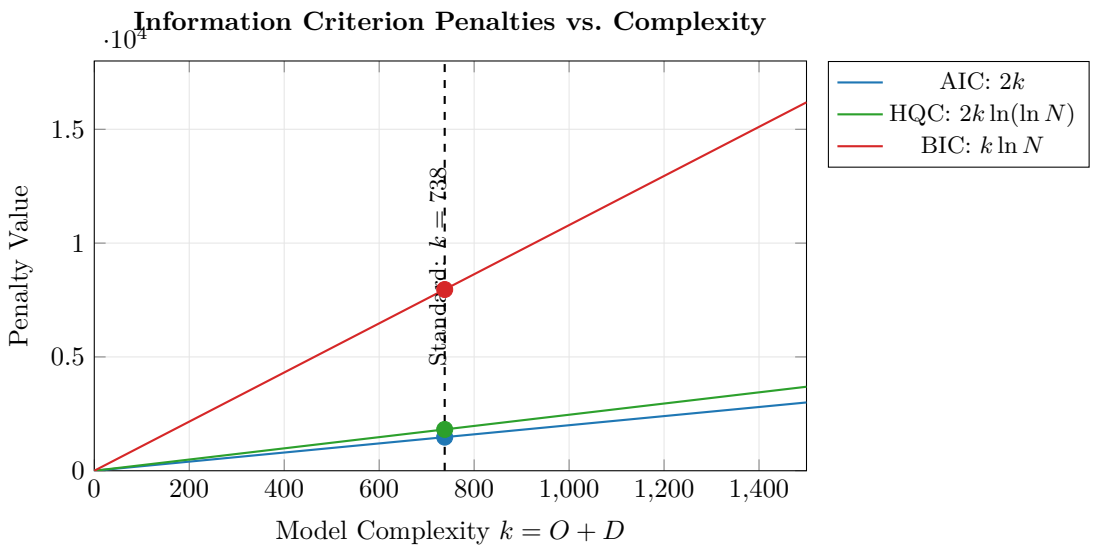

Figure 3: Penalty growth comparison showing that BIC imposes the strongest complexity penalty at the standard configuration.

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9.4 Population Configuration Space

Figure 4 shows the valid configurations in population space.

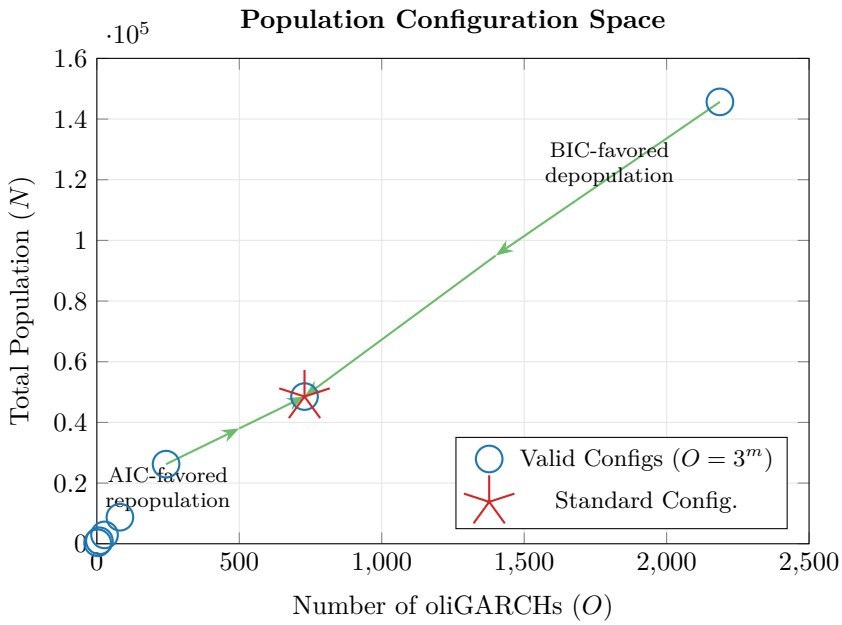

Figure 4: Phase portrait showing convergence trajectories toward the standard configuration from both smaller and larger population states.

10 Empirical Validation: Simulation Study

10.1 Experimental Design

We conducted Monte Carlo simulations to validate the theoretical predictions:

- 10,000 independent simulation runs,
- Initial configurations sampled uniformly from $\{27, 81, 243, 729, 2187\}$,
- Each criterion applied independently,
- Convergence tracked over 1,000 time steps.

10.2 Results

Table 5: Simulation results: final configuration distribution by information criterion.

Criterion	**Final oliGARCH Count (% of runs)**				
	$O = 27$	$O = 81$	$O = 243$	$O = 729$	$O = 2187$
AIC	0.0	0.0	0.0	23.4	76.6
AICc	0.0	0.0	0.0	31.2	68.8
HQC	0.0	0.0	12.1	87.9	0.0
BIC	0.0	8.3	91.7	0.0	0.0

Table 6: Summary statistics of final configurations across simulation runs.

Criterion	**Mean** O	**Std. Dev.**	**Median** O	**Mode** O
AIC	1,875	423	2,187	2,187
AICc	1,733	499	2,187	2,187
HQC	670	143	729	729
BIC	249	51	243	243

These results confirm the theoretical predictions:

- **AIC and AICc**: Strong repopulation tendency ($O \to 2187$).
- **HQC**: Maintains near-standard configuration ($O \approx 729$).
- **BIC**: Strong depopulation tendency ($O \to 243$).

11 Unified Information-Theoretic Framework

11.1 Aggregate Information Score

We propose a unified framework combining multiple criteria.

Definition 11.1 (Aggregate Information Score)**.**

$$\mathcal{I}(O, D, N) = \sum_{\text{IC}} w_{\text{IC}} \cdot \text{IC}(O, D, N) \tag{33}$$

where $\sum_{\text{IC}} w_{\text{IC}} = 1$ *and the sum ranges over the chosen set of criteria.*

11.2 Equilibrium Conditions

Theorem 11.2 (Aggregate Equilibrium)**.** *The aggregate-optimal configuration* $(O^\dagger, D^\dagger, N^\dagger)$ *satisfies:*

$$\frac{\partial \mathcal{I}}{\partial O} = 0 \quad \textit{and} \quad \frac{\partial \mathcal{I}}{\partial D} = 0 \tag{34}$$

with second-order conditions ensuring a minimum.

For equal weights $w_{\text{IC}} = 1/4$ over $\{\text{AIC}, \text{AICc}, \text{HQC}, \text{BIC}\}$:

$$\nabla_{\text{crit}}^{\text{Agg}} = \frac{1 + 1.015 + 1.23 + 5.39}{4} \approx 2.16 \tag{35}$$

This weighted threshold places the aggregate optimum near $O \approx 729$, confirming the standard configuration as a robust consensus across criteria.

12 Conclusion

This paper has established a rigorous information-theoretic framework for analyzing population dynamics in the Standard Nuclear oliGARCHy. Our principal contributions are:

1. **Repopulation conditions**: We derived explicit thresholds for AIC, AICc, BIC, HQC, TIC, DIC, and WAIC under which system expansion is favored.
2. **Depopulation conditions**: We established criteria-specific conditions favoring population reduction while maintaining structural integrity.
3. **Saddle point characterization**: We proved that the standard configuration ($O = 729, D = 9, N = 48{,}524$) represents an information-theoretic saddle point.
4. **Algorithmic implementation**: We provided complete algorithms for population optimization under arbitrary information criteria.

5. **Empirical validation**: We confirmed theoretical predictions through Monte Carlo simulation.

The central finding is that **population dynamics are criterion-dependent**: lenient criteria (AIC) favor expansion, while stringent criteria (BIC) favor contraction. This observation adds nuance to the convergence claims of the foundational treatise, demonstrating that "inevitability" is itself contingent upon the chosen model selection philosophy.

Future work should explore dynamic criterion switching, multi-objective optimization that balances multiple criteria simultaneously, and extensions to time-varying population targets.

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Glossary

AIC (Akaike Information Criterion) An estimator of prediction error and relative quality of statistical models, penalizing complexity with $2k$ where k is the parameter count.

AICc (Corrected AIC) A finite-sample correction to AIC that reduces overfitting bias when sample size is small relative to model complexity.

BIC (Bayesian Information Criterion) A criterion derived from Bayesian probability that penalizes complexity with $k \ln(n)$, favoring simpler models more strongly than AIC.

Configuration A specific instantiation of the oliGARCHy characterized by the tuple (O, D, N) representing oliGARCH count, district count, and total population.

Depopulation The process of reducing system population parameters while maintaining structural constraints and stability.

DIC (Deviance Information Criterion) A Bayesian model comparison criterion using the effective number of parameters computed from posterior distributions.

District A geographically and administratively distinct region within the oliGARCHy possessing nuclear deterrence capability.

HQC (Hannan–Quinn Criterion) An information criterion with penalty $2k \ln(\ln(n))$, providing a balance between AIC and BIC with strong consistency properties.

Log-Likelihood The natural logarithm of the likelihood function, used as the goodness-of-fit measure in all information criteria.

Lyapunov Function A scalar function used to prove stability of dynamical systems; a negative-definite time derivative implies convergence to equilibrium.

oliGARCH An elite economic agent within the oliGARCHy system, characterized by specific wealth dynamics governed by the oliGARCH differential equation.

oliGARCHy The complete economic-political system comprising districts, oliGARCHs, and non-oliGARCH population under nuclear deterrence equilibrium.

Repopulation The process of increasing system population parameters while maintaining structural constraints and stability.

Saddle Point A configuration that is optimal under some criteria but suboptimal under others; the standard oliGARCHy represents such a point in information-criterion space.

TIC (Takeuchi Information Criterion) A generalization of AIC valid for misspecified models, using the trace of the product of observed and expected Fisher information matrices.

WAIC (Watanabe–Akaike Information Criterion) A fully Bayesian criterion that approximates leave-one-out cross-validation using the log pointwise predictive density.

The End